

TCC70xx MCU BSP

API Specification for Windowed Watchdog Timer

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Preliminary version

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1 INTRODUCTION

This document describes the windowed watchdog timer device driver (hereinafter referred to as WDT driver) of TCC70xx MCU BSP. For more detailed information on windowed watchdog timer in the TCC70xx MCU Sub-system, refer to Chapter 20 Windowed Watchdog Timer (WWDT) in "*TCC70xx Full Specification*". [1]

2 TERMS AND ABBREVIATIONS

Table 2.1 Terms and Abbreviation

WWDT	Windowed Watchdog Timer
-------------	-------------------------

3 SOURCE FILES

3.1 Location of Source Files

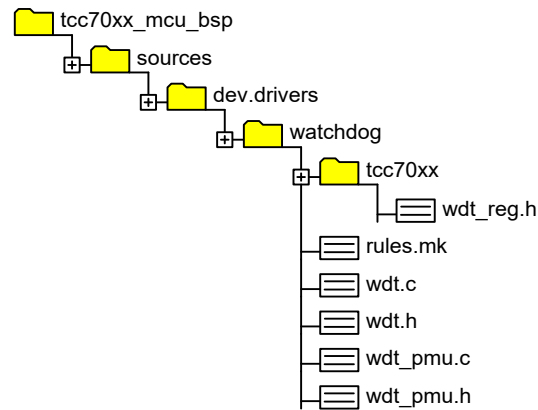


Figure 3.1 Location of Source Files

3.2 Simple Description of Source Files

- wdt_reg.h
 - Device register address and writing password
- rules.mk
 - Rules used for build
- wdt.c
 - WDT driver source
- wdt.h
 - WDT driver API header
- wdt_pmu.c
 - PMU WDT driver source
- wdt_pmu.h
 - PMU WDT driver API header

4 ENUMERATIONS

4.1 WDTResetType_t

Enumeration variables to select whether handshake mode with FMU is enabled or not

Declaration

```
typedef enum WDTResetType
{
    WDT_RST_DIRECT      = 0x0U,
    WDT_RST_FMU_HANDLER = 0x4U
} WDTResetType_t;
```

Description

Value	Description
WDT_RST_DIRECT	System is reset without any interrupting.
WDT_RST_FMU_HANDLER	Activated RST_REQ signal is treated by Fault Management Unit (FMU). You can handle the process after reset signal.

Remark

None

4.2 WDT2oo3Type_t

Enumeration variables to select one of RST counter timeout signals or filtered signal which is 2 out of 3 voted

Declaration

```
typedef enum WDT2003Type
{
    WDT_2003_VOTE = 0x0U,
    WDT_ZERO_0    = 0x1U,
    WDT_ZERO_1    = 0x2U,
    WDT_ZERO_2    = 0x3U
} WDT2oo3Type_t;
```

Description

Value	Description
WDT_2003_VOTE	Watchdog interrupt and watchdog reset request signal select 2 out of 3 vote applied signal.
WDT_ZERO_0	IRQ_CNT_ZERO_0 and RST_CNT_ZERO_0
WDT_ZERO_1	IRQ_CNT_ZERO_1 and RST_CNT_ZERO_1
WDT_ZERO_2	IRQ_CNT_ZERO_2 and RST_CNT_ZERO_2

Remark

None

4.3 WDTErrorCode_t

Enumeration variables for execution error codes of Watchdog timer user interface API

Declaration

```
enum
{
    WDT_ERR_ALREADY_DONE      = (uint32)SAL_DRVID_WDT,
    WDT_ERR_NOT_INITIALIZED   = (uint32)SAL_DRVID_WDT + 1UL,
    WDT_ERR_INVALID_PARAM     = (uint32)SAL_DRVID_WDT + 2UL,
    WDT_ERR_NOT_SUPPORT_MODE  = (uint32)SAL_DRVID_WDT + 3UL
}; //WDTErrorCode_t
```

Description

Value	Description
WDT_ERR_ALREADY_DONE	A job is re-executed.
WDT_ERR_NOT_INITIALIZED	WDT module is <u>not</u> initialized.
WDT_ERR_INVALID_PARAM	Invalid parameters are passed.
WDT_ERR_NOT_SUPPORT_MODE	Mode is <u>not</u> supported.

Remark

None

4.4 WDTPMURestType_t

Enumeration variables to select whether handshake mode with FMU is enabled or not

Declaration

```
typedef enum WDTPMURestType
{
    WDTPMU_RST_DIRECT          = 0x0U,
    WDTPMU_RST_FMU_HANDLER     = 0x4U
} WDTPMURestType_t;
```

Description

Value	Description
WDTPMU_RST_DIRECT	System is reset without any interrupting.
WDTPMU_RST_FMU_HANDLER	Activated PMU Watchdog RST_REQ signal is treated by Fault Management Unit (FMU). You can handle the process after reset signal.

Remark

None

4.5 WDTPMU2oo3Type_t

Enumeration variables to select one of PMU WDT's RST counter timeout signals or filtered signal which is 2 out of 3 voted

Declaration

```
typedef enum WDTPMU2003Type
{
    WDTPMU_2003_VOTE    = 0x0U,
    WDTPMU_ZERO_0       = 0x1U,
    WDTPMU_ZERO_1       = 0x2U,
    WDTPMU_ZERO_2       = 0x3U
} WDTPMU2oo3Type_t;
```

Description

Value	Description
WDTPMU_2003_VOTE	PMU Watchdog interrupt and PMU watchdog reset request signal select 2 out of 3 vote applied signal.
WDTPMU_ZERO_0	IRQ_CNT_ZERO_0 and RST_CNT_ZERO_0
WDTPMU_ZERO_1	IRQ_CNT_ZERO_1 and RST_CNT_ZERO_1
WDTPMU_ZERO_2	IRQ_CNT_ZERO_2 and RST_CNT_ZERO_2

Remark

None

4.6 WDTPMUErrorCode_t

Enumeration variables for execution error codes of PMU Watchdog timer user interface API

Declaration

```
enum
{
    WDTPMU_ERR_ALREADY_DONE           = (uint32)SAL_DRVID_WDT + 10UL,
    WDTPMU_ERR_NOT_INITIALIZED        = (uint32)SAL_DRVID_WDT + 11UL,
    WDTPMU_ERR_INVALID_PARAM          = (uint32)SAL_DRVID_WDT + 12UL,
    WDTPMU_ERR_NOT_SUPPORT_MODE       = (uint32)SAL_DRVID_WDT + 13UL
}; //WDTPMUErrorCode_t
```

Description

Value	Description
WDTPMU_ERR_ALREADY_DONE	A job is re-executed.
WDTPMU_ERR_NOT_INITIALIZED	WDT module is <u>not</u> initialized.
WDTPMU_ERR_INVALID_PARAM	Invalid parameters are passed.
WDTPMU_ERR_NOT_SUPPORT_MODE	Mode is <u>not</u> supported.

Remark

None

5 CONSTANTS

5.1 Function Indexes

Indexing constants correspond to each API function. These indexes are used for debugging or for informing information of WDT API function.

Declaration

```
#define WDT_API_COMMON_INDEX      (0x0)
#define WDT_API_STOP              (WDT_API_COMMON_INDEX + 0)
#define WDT_API_SET_SM_MODE      (WDT_API_COMMON_INDEX + 1)
#define WDT_API_KICK_ACT         (WDT_API_COMMON_INDEX + 2)
#define WDT_API_RESET_SYS        (WDT_API_COMMON_INDEX + 3)
#define WDT_API_INIT              (WDT_API_COMMON_INDEX + 4)

#define WDTPMU_API_COMMON_INDEX  (0x0)
#define WDTPMU_API_STOP          (WDTPMU_API_COMMON_INDEX + 0)
#define WDTPMU_API_SET_SM_MODE  (WDTPMU_API_COMMON_INDEX + 1)
#define WDTPMU_API_KICK_ACT     (WDTPMU_API_COMMON_INDEX + 2)
#define WDTPMU_API_RESET_SYS    (WDTPMU_API_COMMON_INDEX + 3)
#define WDTPMU_API_INIT          (WDTPMU_API_COMMON_INDEX + 4)
```

Description

Value	Description
WDT_API_COMMON_INDEX	Starting index of Watchdog timer module
WDT_API_STOP	Index of <i>WDT_Stop()</i> function
WDT_API_SET_SM_MODE	Index of <i>WDT_SmMode()</i> function
WDT_API_KICK_ACT	Index of <i>WDT_KickPing()</i> function
WDT_API_RESET_SYS	Index of <i>WDT_ResetSystem()</i> function
WDT_API_INIT	Index of <i>WDT_Init()</i> function
WDTPMU_API_COMMON_INDEX	Starting index of PMU Watchdog timer module
WDTPMU_API_STOP	Index of <i>WDTPMU_Stop()</i> function
WDTPMU_API_SET_SM_MODE	Index of <i>WDTPMU_SmMode()</i> function
WDTPMU_API_KICK_ACT	Index of <i>WDTPMU_KickPing()</i> function
WDTPMU_API_RESET_SYS	Index of <i>WDTPMU_ResetSystem()</i> function
WDTPMU_API_INIT	Index of <i>WDTPMU_Init()</i> function

Remark

None

6 APIs

6.1 WDT_KickPing

Processes kick action of WDT counter as well as clears and resets WDT IRQ counter.

Declaration

```
void WDT_KickPing  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

6.2 WDT_ResetSystem

Resets or reboots the system right away.

Declaration

```
void WDT_ResetSystem  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

6.3 WDT_Stop

Stops WDT counter and unregisters an interrupt handler.

Declaration

```
void WDT_Stop  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

6.4 WDT_SmMode

Sets the configuration of safety mechanism on WDT device.

Declaration

```
SALRetCode_t WDT_SmMode
(
    boolean          ucUseFmu,
    WDTResetType_t   uiUseHandler,
    WDT2oo3Type_t     uiVote
);
```

Parameters

Value	Description
<i>ucUseFmu</i>	<i>[in] Flag to use handshake with FMU. If true, handshake mode will be enabled.</i>
<i>uiUseHandler</i>	<i>[in] When handshake with FMU, external FMU handler is available. Refer to WDTResetType_t.</i>
<i>uiVote</i>	<i>[in] WDT reset request selection mode, selecting one of 3 reset request line or 2 out of 3 voted reset requests. Refer to WDT2oo3Type_t.</i>

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

6.5 WDT_Init

Initializes configuration registers, registers an interrupt handler for kick action, and starts WDT counter.

Declaration

```
SALRetCode_t WDT_Init  
(  
    void  
);
```

Parameters

None

Return

<i>Value</i>	<i>Description</i>
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

6.6 WDTPMU_KickPing

Processes kick action of PMU WDT counter as well as clears and resets PMU WDT IRQ counter.

Declaration

```
void WDTPMU_KickPing  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

6.7 WDTPMU_ResetSystem

Resets or reboots the system right away.

Declaration

```
void WDTPMU_ResetSystem  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

6.8 WDTPMU_Stop

Stops PMU WDT counter and unregisters an interrupt handler.

Declaration

```
void WDTPMU_Stop  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

6.9 WDTPMU_SmMode

Sets the configuration of safety mechanism on PMU WDT device.

Declaration

```
SALRetCode_t WDTPMU_SmMode
(
    boolean          ucUseFmu,
    WDTPMUResetType_t uiUseHandler,
    WDTPMU2oo3Type_t uiVote
);
```

Parameters

Value	Description
<i>ucUseFmu</i>	<i>[in] Flag to use handshake with FMU. If true, handshake mode will be enabled.</i>
<i>uiUseHandler</i>	<i>[in] When handshake with FMU, external FMU handler is available. Refer to WDTPMUResetType_t.</i>
<i>uiVote</i>	<i>[in] PMU WDT reset request selection mode, selecting one of 3 reset request line or 2 out of 3 voted reset requests. Refer to WDTPMU2oo3Type_t.</i>

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

6.10 WDTPMU_Init

Initializes configuration registers, registers an interrupt handler for kick action, and starts PMU WDT counter.

Declaration

```
SALRetCode_t WDTPMU_Init
(
    void
);
```

Parameters

None

Return

Value	Description
SAL_RET_SUCCESS	Success
SAL_RET_FAILED	Failure

Remark

None

7 HOW TO USE WDT DRIVER

7.1 WDT Default Operation

The normal operation of Watchdog is turned on by MCU system via ***WDT_Init()***. If the operation of Watchdog is off via ***WDT_Stop()***, the MCU system reboots after configuring the reset counter value.

```
{
    SALRetCode_t ret;

    ret = WDT_Init();

    if (ret == SAL_RET_SUCCESS)
    {
        ...
    }
}
```

```
{
    WDT_Stop();
}
```

7.2 WDT Customized Operation

You can customize the operation by using Watchdog driver and Timer driver. You should call the ***WDT_KickPing()*** for MCU system to be alive.

```
#define WDT_IRQ_TIME          (10UL)           // 10 seconds
#define WDT_RESET_TIME       (24UL)           // 24 seconds
#define TIMER_TESTING_PRERIGHT_TIME (10 * 1000 * 1000) // 10 sec

{
    SALRetCode_t ret;

    (void)WDT_SmMode(TRUE, WDT_RST_DIRECT, WDT_2003_VOTE);

    ret = WDT_SetTimeout(WDT_RESET_TIME, WDT_IRQ_TIME);

    if (ret == SAL_RET_SUCCESS)
    {
        (void)TIMER_EnableWithMode(TIMER_CH_1,
                                    TIMER_TESTING_PRERIGHT_TIME,
                                    TIMER_OP_FREERUN,
                                    TIMER_Handler,
                                    NULL_PTR);

        GIC_IntSrcEn(GIC_TIMER_0 + TIMER_CH_1);

        WDT_Start();
    }
}
```

```
sint32 TIMER_Handler(TIMERChannel_t iChannel, void * pArgs)
{
    WDT_KickPing();

    return SAL_RET_SUCCESS;
}
```

7.3 PMU WDT Default Operation

The normal operation of PMU Watchdog is turned on by MCU system via *WDTPMU_Init()*. If the operation of PMU Watchdog is off via *WDTPMU_Stop()*, the MCU system reboots after configuring the reset counter value.

```
{  
    SALRetCode_t ret;  
  
    ret = WDTPMU_Init();  
  
    if (ret == SAL_RET_SUCCESS)  
    {  
        ...  
    }  
}
```

```
{  
    WDTPMU_Stop();  
}
```


7.4 PMU WDT Customized Operation

You can customize the operation by using PMU Watchdog driver and Timer driver. You should call the *WDTPMU_KickPing()* for MCU system to be alive.

```
#define WDTPMU_IRQ_TIME          (10UL)          // 10 seconds
#define WDTPMU_RESET_TIME       (24UL)          // 24 seconds
#define TIMER_TESTING_PRERIGHT_TIME (10 * 1000 * 1000) // 10 sec

{
    SALRetCode_t ret;

    (void)WDTPMU_SmMode(TRUE, WDTPMU_RST_DIRECT, WDTPMU_2003_VOTE);

    ret = WDTPMU_SetTimeout(WDTPMU_RESET_TIME, WDTPMU_IRQ_TIME);

    if (ret == SAL_RET_SUCCESS)
    {
        (void)TIMER_EnableWithMode(TIMER_CH_1,
                                    TIMER_TESTING_PRERIGHT_TIME,
                                    TIMER_OP_FREERUN,
                                    TIMER_Handler,
                                    NULL_PTR);

        GIC_IntSrcEn(GIC_TIMER_0 + TIMER_CH_1);

        WDTPMU_Start();
    }
}
```

```
sint32 TIMER_Handler(TIMERChannel_t iChannel, void * pArgs)
{
    WDTPMU_KickPing();

    return SAL_RET_SUCCESS;
}
```

8 REFERENCES

- [1] TCC70xx Full Specification
- [2] Contact Telechips for more details: sales@telechips.com

9 REVISION HISTORY

Rev. 0.10: 2021-09-03

- Preliminary version release

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